Skills Summary

Using Normal-Model z-Scores and Percentiles

Use this reference while completing your worksheet or when studying.

Skill 1 — Standardizing: from a value to z

Formula: $z = \dfrac{x-\mu}{\sigma}$ — how many standard deviations the value sits above (positive) or below (negative) the mean. Subtract first, divide second. A z-score carries no units, which is exactly what makes it comparable across different distributions.

Example: A distribution has $\mu = 65$ and $\sigma = 6$. Find the z-score of 74.
Step 1. The question asks how far above average the value is in standard-deviation units, so standardize.
Step 2. $z = \dfrac{74-65}{6} = \dfrac{9}{6}$.
$\Rightarrow$ $z = \mathbf{\frac{3}{2}}$, that is, 1.5 standard deviations above the mean.

Try it: Same $\mu = 65$, $\sigma = 6$. Find the z-score of 58.

check: $z = -\mathbf{\frac{7}{6}}$

Skill 2 — Unstandardizing: from z back to a value

Formula: $x = \mu + z\sigma$. Read it as "start at the mean, then walk z standard deviations". A negative z walks downward. This is the move that answers every "find the score at the kth percentile" question once you know the z for that percentile.

Example: $\mu = 200$, $\sigma = 25$. Find the value with $z = 1.2$.
Step 1. We are given the position in standard-deviation units and want the raw value, so run the formula backwards.
Step 2. $x = 200 + 1.2(25) = 200 + 30$.
$\Rightarrow$ $x = \mathbf{230}$

Try it: Same $\mu = 200$, $\sigma = 25$. Find the value with $z = -0.8$.

check: $x = \mathbf{180}$

Skill 3 — Percentiles from the empirical rule

Rule: in a normal model about 68% of values lie within 1σ of the mean, 95% within 2σ, and 99.7% within 3σ. The curve is symmetric, so half of the remainder sits in each tail. To get a percentile, add 50 (everything below the mean) to half of the central percentage — or subtract, for a point below the mean.

Example: What percentage of a normal distribution lies below $z = 2$?
Step 1. The rule gives a symmetric central band, so split it in half and add the half of the curve below the mean.
Step 2. $50 + \dfrac{95}{2} = 50 + 47.5$.
$\Rightarrow$ **97.5** percent, so $z = 2$ marks the 97.5th percentile.

Try it: What percentage lies below $z = -2$?

check: **2.5** percent

Skill 4 — Comparing across different distributions

Rule: two raw values from different distributions cannot be compared directly — they are measured against different means and different spreads. Standardize each against *its own* μ and σ, then compare the z-scores. The larger z is the more extreme result relative to its own group, whatever the raw numbers say.

Example: A time of 33 comes from a group with $\mu = 30$, $\sigma = 2$. Find its z-score.
Step 1. Before any comparison can be made, each value has to be measured against its own group, so standardize this one first.
Step 2. $z = \dfrac{33 - 30}{2}$.
$\Rightarrow$ $z = \mathbf{\frac{3}{2}}$, which can now be set beside any other group's z.

Try it: A score of 62 comes from a group with $\mu = 50$, $\sigma = 10$. Find its z-score.

check: $z = \mathbf{\frac{6}{5}}$